

Python Programming Language Overview

Python is a high-level, interpreted programming language known for its readability and clean syntax. It uses dynamic typing and automatic garbage collection.

Python was created by Guido van Rossum and first released in 1991.

Key Features of Python:

1. Interpreted: Code is executed line by line.
2. Multi-paradigm: Supports object-oriented, functional, and procedural styles.
3. Extensive standard library: Batteries included philosophy.

Python Data Structures and Frameworks

Core data structures include lists, tuples, dictionaries, and sets.

Dictionaries are key-value mapping stores with $O(1)$ average lookup time.

Popular Python web frameworks include FastAPI, Flask, and Django.

FastAPI is modern and fast, built on Starlette and Pydantic.